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Assignment # 3

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Functional Dependencies

Q1

$R(A, B, C, D, E)$

FDs $\{C \rightarrow AB, A \rightarrow E, D \rightarrow E, BD \rightarrow C, CD \rightarrow B\}$

Candidate Keys: - $AB \not\subseteq D \not\subseteq E$

$CD^+ = \{DECAB\}$

$BD^+ = \{DBC A E\}$

CD and BD are the candidate keys.

Q2

$R(A, B, C, D)$

FDs $\{AB \rightarrow C, CD \rightarrow B, AD \rightarrow B, AC \rightarrow D\}$

Candidate keys: - $AB \not\subseteq D \not\subseteq C$

$BA^+ = \{ABCD\}$

$DA^+ = \{ADBC\}$

$CA^+ = \{ACDB\}$

AB, AC, AD are the candidate keys.

Q3

$R(A, B, C, D, E)$

FDs $\{A \rightarrow C, C \rightarrow BD, D \rightarrow A\}$

Candidate keys: - $AB \not\subseteq D \not\subseteq E$

$AE^+ = \{AECBD\}$

$DE^+ = \{DEACB\}$

$CE^+ = \{CEBDA\}$

AE, CE and DE are the candidate keys.

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Q4

R(A, B, C, D, E, G)

FDs:-

$$F_1 \{ A \rightarrow C, AB \rightarrow C, C \rightarrow DG, CD \rightarrow G, \\ EC \rightarrow AB, EG \rightarrow C \}$$

$$F_2 \{ A \rightarrow C, C \rightarrow D, C \rightarrow G, EC \rightarrow A, \\ EC \rightarrow B, EG \rightarrow C \}$$

Checking if all F_1 FDs can be derived from F_2 FDs.

- $F_1 \{ A \rightarrow C \} = F_2 \{ A \rightarrow C \}$
- $F_1 \{ AB \rightarrow C \} = F_2 \{ AB \rightarrow C \}$ as $F_2 \{ A \rightarrow C \}$
- $F_1 \{ C \rightarrow DG \} = F_2 \{ C \rightarrow DG \}$ as $F_2 \{ C \rightarrow D, C \rightarrow G \}$
- $F_1 \{ CD \rightarrow G \} = F_2 \{ CD \rightarrow G \}$ as $F_2 \{ C \rightarrow G \}$
- $F_1 \{ EC \rightarrow AB \} = F_2 \{ EC \rightarrow AB \}$ as $F_2 \{ EC \rightarrow A, EC \rightarrow B \}$
- $F_1 \{ EG \rightarrow C \} = F_2 \{ EG \rightarrow C \}$ as $F_2 \{ EG \rightarrow C \}$

All F_2 functional dependencies can be created from function 1 functional dependencies.

Checking if all F_2 FDs can be derived from F_1 FDs.

- $F_2 \{ A \rightarrow C \} = F_1 \{ A \rightarrow C \}$ as $F_1 \{ A \rightarrow C \}$
- $F_2 \{ C \rightarrow D \} = F_1 \{ C \rightarrow D \}$ as $F_1 \{ C \rightarrow DG \}$
- $F_2 \{ C \rightarrow G \} = F_1 \{ C \rightarrow G \}$ as $F_1 \{ C \rightarrow DG \}$
- $F_2 \{ EC \rightarrow A \} = F_1 \{ EC \rightarrow A \}$ as $F_1 \{ EC \rightarrow AB \}$
- $F_2 \{ EC \rightarrow B \} = F_1 \{ EC \rightarrow B \}$ as $F_1 \{ EC \rightarrow AB \}$
- $F_2 \{ EG \rightarrow C \} = F_1 \{ EG \rightarrow C \}$ as $F_1 \{ EG \rightarrow C \}$

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All F_1 functional dependencies can be created from F_2 functional dependencies.

Since all functional dependencies in F_1 can be derived from F_2 and vice versa, we can conclude that F_1 and F_2 are equivalent sets of functional dependencies.

Q5

$R(A, B, C, D, E, G)$

$FDs = \{ ABC \rightarrow CDEG, C \rightarrow E, A \rightarrow B, D \rightarrow G \}$

Minimal Cover:-

Step 1 Check the R.H.S

$\begin{matrix} \nearrow R & & \nearrow R & & \nearrow R \\ ABC \rightarrow C, & ABC \rightarrow D, & ABC \rightarrow E, & ABC \rightarrow G, \\ C \rightarrow E, & A \rightarrow B, & D \rightarrow G \end{matrix}$

Step 2 Check Redundant

$ABC \rightarrow D, C \rightarrow E, A \rightarrow B, D \rightarrow G$

Step 3 Check the L.H.S

$A^+ = \{ A, B \}$

$B^+ = \{ B \}$

$C^+ = \{ C, E \}$

$AC \rightarrow D, C \rightarrow E, A \rightarrow B, D \rightarrow G$

Possible keys:- $ABCD \notin K$

$AC^+ = \{ ACB, G, D, E \}$

AC^+ is the ~~candidate~~ minimal super key.

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Q6

$R(A, B, C, D, E, G)$

FDs ($D \rightarrow E$, $ABC \rightarrow BDE$, $B \rightarrow G$, $A \rightarrow C$,
 $ABC \rightarrow G$)

Minimal Cover

Step 1 Look at the R.H.S

$D \rightarrow E$, $ABC \rightarrow B$, $ABC \rightarrow D$, $ABC \rightarrow E$,
 $B \rightarrow G$, $A \rightarrow C$, $ABC \rightarrow G$

Step 2 Check Redundant

$D \rightarrow E$, $B \rightarrow G$, ~~$ABC \rightarrow B$~~ , $ABC \rightarrow D$, $A \rightarrow C$,

Step 3 check at the L.H.S

$D \rightarrow E$, $B \rightarrow G$, ~~$ABC \rightarrow B$~~ , $AB \rightarrow D$, $A \rightarrow C$

$A^+ = \{AC\}$

$B^+ = \{BG\}$

$C^+ = \{C\}$

Possible keys: A ~~B~~ ~~$\emptyset$~~ ~~G~~

$AB^+ = \{AGCBDE\}$

Minimal super key is AB^+ .

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Normalization for Relational Databases

a

$R(A, B, C, D, E, F, G)$

FDs ($A \rightarrow G, F \rightarrow E, G \rightarrow DB, D \rightarrow C$)

a

Possible Keys :- $AB \cancel{C} \cancel{D} \cancel{E} F \cancel{G}$

$AF^+ = \{A B C D E F G\}$

AF^+ is the only candidate key.

b

	$A \rightarrow G$	$F \rightarrow E$	$G \rightarrow DB$	$D \rightarrow C$
BCNF	X	X	X	X
3NF	X	X	X	X
2NF	X	X	✓	✓
1NF	✓	✓	✓	✓

1NF is the best normal form that satisfies R.

c

$R(A, B, C, D, E, F, G) - R_4(A, F, G)$

$C_k: AFG^+$

$R_1(A, G)$

$R_2(F, E)$

$R_3(B, G, D, C)$

FD ($A \rightarrow G$)

FD ($F \rightarrow E$)

FDs ($G \rightarrow DB, D \rightarrow C$)

$C_k: A$

$C_k: F$

$C_k: G^+$

d

Only $D \rightarrow C$ is not in 3NF.

$R_3(B, C, D, G)$

$R_5(B, D, G)$

$R_6(D \rightarrow C)$

FD ($G \rightarrow DB$)

FD ($D \rightarrow C$)

$C_k: G^+$

$C_k: D^+$

e

All relations are in BCNF.

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Following are the BCNF schema relations:

	<u>Keys</u>	<u>FDs</u>
R1 (A, G)	A^+	$A \rightarrow G$
R2 (E, F)	F^+	$F \rightarrow E$
R5 (B, D, G)	G^+	$G \rightarrow DB$
R6 (D, C)	D^+	$D \rightarrow C$
R7 (A, F, G, D)	$AFGD^+$	$\emptyset$

All dependencies are preserved.

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Q2

$R(A, B, C, D, E)$

$FDs \{A \rightarrow BC, C \rightarrow D, E \rightarrow D, BE \rightarrow A\}$

$CK: ABE \not\subseteq D$

$BE^+ = \{BEACD\}$

$AE^+ = \{AEB CD\}$

BE^+ and AE^+ are the candidate keys.

a

	$A \rightarrow BC$	$C \rightarrow D$	$E \rightarrow D$	$BE \rightarrow A$
BCNF	$\times$	$\times$	$\times$	$\checkmark$
3NF	$\times$	$\times$	$\times$	$\checkmark$
2NF	$\checkmark$	$\checkmark$	$\times$	$\checkmark$
1NF	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

b

1NF is the best normal form that satisfies R.

$R(A, B, C, D, E)$

$R_1(E, D)$

$FD \{E \rightarrow D\}$

$CK: E^+$

$R_2(A, B, C, E, D)$

$FDs \{A \rightarrow BC, BE \rightarrow A, C \rightarrow D\}$

$CK: BE^+ \text{ and } AE^+$

$ABD \not\subseteq E$

All 2NF violations have been handled.

c

No $A \rightarrow BC$ and $C \rightarrow D$ are not in 3NF.

$R_2(A, B, C, D, E)$

$R_3(A, B, C)$

$FD \{A \rightarrow BC\}$

$CK: A^+$

$R_4(A, B, E)$

$FD \{BE \rightarrow A\}$

$CK: BE^+$

$R_5(C, D)$

$FD \{C \rightarrow D\}$

$CK: C^+$

All 3NF violations have been handled.

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d

All the relations are in BCNF.

Following are the BCNF schema relations:-

	<u>Keys</u>	<u>FDs</u>
$R_1(E, D)$	E^+	$E \rightarrow D$
$R_3(A, B, C)$	A^+	$A \rightarrow BC$
$R_4(A, B, E)$	BE^+	$BE \rightarrow A$
$R_6(C, D)$	C^+	$C \rightarrow D$

All dependencies were preserved.

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Q3

$R(A, B, C, D)$

FDs $(B \rightarrow D, D \rightarrow A)$

	$B \rightarrow D$	$D \rightarrow A$
BCNF	$\times$	$\times$
3NF	$\times$	$\times$
2NF	$\times$	$\checkmark$
1NF	$\checkmark$	$\checkmark$

$C_k: ABCD$

$BC^+ = \{BCDA\}$

1NF is the best normal form that satisfies R.

$R(A, B, C, D)$

$R_1(B, D, A)$

FDs $(B \rightarrow D, D \rightarrow A)$

$C_k: B^+$

$R_2(C, B)$

FDs $(\emptyset)$

$C_k: CB^+$

2NF has been achieved for R1.

$R_1(B, D, A)$

$R_3(B, D)$

FD $(B \rightarrow D)$

$C_k: B^+$

$R_4(D, A)$

FD $(D \rightarrow A)$

$C_k: D^+$

3NF and BCNF has been achieved.

Following are the BCNF schema relations.

	Keys	FDs
$R_2(C, B)$	CB^+	$\emptyset$
$R_3(B, D)$	B^+	$B \rightarrow D$
$R_4(D, A)$	D^+	$D \rightarrow A$

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All dependencies were preserved.

Q4 $R(A, B, C, D, E)$ FD, $(A \rightarrow BC, C \rightarrow D, E \rightarrow D, BE \rightarrow A)$ a $R_1(A, E)$ $R_2(A, B, C)$ $R_3(D, E)$ $R_1 \bowtie R_2 = (A, B, C, E)$ Finding the closure of the common attribute of R_1 and R_2 that is A. $A^+ = \{A, B, C, D\}$

Since A^+ is a candidate key in ~~R_2~~ R_2 ,
 now we will check for the common attribute of $R_1 \bowtie R_2$ and R_3 that is E.

 $E^+ = \{E, D\}$

Since E^+ is a candidate key in R_3 ,
 the composition is lossless decomposition.

Now checking is it dependency preserving:-

$R_1(A, E)$	$R_2(A, B, C)$	$R_3(D, E)$
$A \rightarrow E$ X	$A \rightarrow BC$ ✓	$D \rightarrow E$ X
$E \rightarrow A$ X	$B \rightarrow AC$ X	$E \rightarrow D$ ✓
	$C \rightarrow AB$ X	
	$BC \rightarrow A$ X	
	$AC \rightarrow B$ ✓	
	$AB \rightarrow C$ ✓	

Dependencies $A \rightarrow BC, AC \rightarrow B, AB \rightarrow C, E \rightarrow D$

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$$A^+ = \{A B C D\}$$

$$E^+ = \{E D\}$$

$$B^+ = \{B\}$$

$$C^+ = \{C D\}$$

$$B C^+ = \{B C D\}$$

$$A C^+ = \{A C D B\}$$

$$A B^+ = \{A B C D\}$$

$$D^+ = \{D\}$$

$$E^+ = \{E D\}$$

Seeing which dependencies are preserved:

$$A \rightarrow B C \quad \checkmark$$

$$C \rightarrow D \quad \times$$

$$E \rightarrow D \quad \checkmark$$

$$B E \rightarrow A \quad \times$$

Hence, the decomposition was lossless but not dependency preserving as the $C \rightarrow D$ and $B E \rightarrow A$ dependencies were lost during the decomposition.

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b
 $R_1(A, E)$

$R_2(A, B, C)$

$R_3(C, D)$

$R_1 \bowtie R_2 = (A, B, C, E)$

Finding the closure of the common attribute of R_1 and R_2 that is A.

$A^+ = \{A, B, C, D\}$

Since A^+ is a candidate key in R_2 , now we will check for the common attribute of $R_1 \bowtie R_2$ and R_3 that is C.

$C^+ = \{C, D\}$

Since C^+ is a candidate key in R_3 , the composition is lossless decomposition.

Now checking is it dependency preserving:-

$R_1(A, E)$

$A \rightarrow E$ X

$E \rightarrow A$ X

$R_2(A, B, C)$

$A \rightarrow BC$ ✓

$B \rightarrow AC$ X

$C \rightarrow AB$ X

$BC \rightarrow A$ X

$AC \rightarrow B$ ✓

$AB \rightarrow C$ ✓

$R_3(C, D)$

$C \rightarrow D$ ✓

$D \rightarrow C$ X

Dependencies:-

$A \rightarrow BC, AC \rightarrow B, AB \rightarrow C, C \rightarrow D$

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~~A¹~~ Seeing which dependencies are preserved:

$A \rightarrow BC$ ✓

$C \rightarrow D$ ✓

$E \rightarrow D$ ✗

$BE \rightarrow A$ ✗

Hence, the decomposition was lossless but not dependency preserving as the $E \rightarrow D$ and $BE \rightarrow A$ dependencies were lost during the decomposition.

C

$R_1(A, B, C)$

$R_2(C, D)$

$R_3(D, E)$

$R_1 \bowtie R_2 = (A, B, C, D)$

Finding the closure of the common attribute of R_1 and R_2 that is C .

$C^+ = \{C, D\}$

Since C^+ is a candidate key in R_2 , now we will check for the common attribute of $R_1 \bowtie R_2$ and R_3 that is D .

$D^+ = \{D\}$

Since D^+ is not a candidate key in any of the relations $R_1 \bowtie R_2$ or R_3 , the composition is lossy decomposition.

Now checking is it dependency preserving:-

$R_1(A, B, C)$ $R_2(C, D)$ $R_3(C, D, E)$

$A \rightarrow BC$ ✓

$C \rightarrow D$ ✓

$D \rightarrow E$ ✗

$B \rightarrow AC$ ✗

$D \rightarrow C$ ✗

$E \rightarrow D$ ✓

$C \rightarrow AB$ ✗

$BC \rightarrow A$ ✗

$AC \rightarrow B$ ✓

$AB \rightarrow C$ ✓

Dependencies:-

$A \rightarrow BC, AC \rightarrow B, AB \rightarrow C, C \rightarrow D, E \rightarrow D$

Seeing which dependencies are preserved

$A \rightarrow BC$ ✓

$C \rightarrow D$ ✓

$E \rightarrow D$ ✓

$BE \rightarrow A$ ✗

$BE^+ = \{BE, D\}$

Hence, the decomposition was lossy and dependencies were not preserved as the $BE \rightarrow A$ dependency was lost during the decomposition.

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Q5

$$F1 = \{ A \rightarrow C, AB \rightarrow C, C \rightarrow D, CD \rightarrow I, \\ EC \rightarrow AB, EI \rightarrow C \}$$

$$F2 = \{ A \rightarrow C, C \rightarrow D, C \rightarrow I, EC \rightarrow A, \\ EC \rightarrow B, EI \rightarrow C \}$$

F1 covers F2:-

$$F2 = \{ A \rightarrow C, C \rightarrow D, C \rightarrow I, EC \rightarrow A, \\ EC \rightarrow B, EI \rightarrow C \}$$

$$A^+ \text{ of } F1 = \{ ACDI \}$$

$$C^+ \text{ of } F1 = \{ CDI \}$$

$$EC^+ \text{ of } F1 = \{ ECDIAB \}$$

$$EI^+ \text{ of } F1 = \{ EICDAB \}$$

F2 covers F1:-

$$F1 = \{ A \rightarrow C, AB \rightarrow C, C \rightarrow D, CD \rightarrow I, \\ EC \rightarrow AB, EI \rightarrow C \}$$

$$A^+ \text{ of } F2 = \{ ACDI \}$$

$$AB^+ \text{ of } F2 = \{ ABCDI \}$$

$$C^+ \text{ of } F2 = \{ CDI \}$$

$$CD^+ \text{ of } F2 = \{ CDI \}$$

$$EC^+ \text{ of } F2 = \{ ECDIAB \}$$

$$EI^+ \text{ of } F2 = \{ EICDAB \}$$

As F1 covers F2 and vice versa,
hence the FDs are equivalent.

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Q6

$R(A, B, C, D, E)$
 $FDs \{A \rightarrow BC, C \rightarrow D, E \rightarrow D, BE \rightarrow A\}$

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$R_1(A, E)$
 $R_2(A, B, C)$
 $R_3(D, E)$

$R_1 \bowtie R_2 = (A, B, C, E)$

Finding the closure of the common attribute of R_1 and R_2 that is A .

$A^+ = \{A, B, C, D\}$

Since A^+ is a candidate key in R_2 , now we will check for the common attribute of $R_1 \bowtie R_2$ and R_3 that is E .

$E^+ = \{E, D\}$

Since E^+ is a candidate key in R_3 , the composition is lossless decomposition.

b

$R_1(A, E)$ $R_2(A, B, C)$ $R_3(C, D)$
 $R_1 \bowtie R_2 = (A, B, C, E)$

Finding the closure of the common attribute of R_1 and R_2 that is A . $A^+ = \{A, B, C, D\}$.

Since A^+ is a candidate key in R_2 , now we will check for the common attributes of $R_1 \bowtie R_2$ and R_3 that is C . $C^+ = \{C, D\}$.

Since C^+ is a candidate key in R_3 , the composition is lossless decomposition.

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